

Verrell's Law — Mathematical Reference (v1.2)

Framework: Verrell's Law

Author: M.R.

Maintainer: Inappropriate Media Limited (t/a Collapse Aware AI)

Status: Public reference — extends CANONICAL_NOTATION_v1.0

Epistemic note: Claims are tagged **DERIVED** **ANSATZ** **SPECULATIVE**. The framework is presented as a set of testable hypotheses and research questions, not established scientific law. Consistent with CANONICAL_NOTATION_v1.0; new structure (§4–§5) is research-stage and explicitly tagged. No canonical symbol is silently altered.

1. Canonical Selection Equation

DERIVED

Memory-weighted selection over candidate states — a softmax with a memory-derived bias added to present-state utility:

$$P(y_i \mid S_t, O_t, M_t) = \frac{\exp(U(y_i; S_t, O_t) + \lambda B(y_i; M_t))}{\sum_j \exp(U(y_j; S_t, O_t) + \lambda B(y_j; M_t))}$$

Symbol	Meaning
y_i	Candidate output / behavioural branch / future state
S_t, O_t, M_t	Current state, current observation/input, retained memory history
$U(y_i; S_t, O_t)$	Present-state utility (memory-independent)
$B(y_i; M_t)$	Memory-derived bias term; encodes Active Information Weight (AIW)
λ	Memory-bias coupling coefficient

Reduction. At $\lambda = 0$ the rule reduces exactly to memoryless softmax selection — the zero-bias baseline against which any memory effect is measured.

3. WEL — Bias Decomposition

ANSATZ

The bias term is a sum over retained traces k :

$$B(y_i; M_t) = \sum_k a_k \phi_k(y_i) w_k(t)$$

Symbol	Meaning
$\phi_k(y_i)$	Alignment of candidate y_i with trace k
a_k	Intrinsic salience of trace k
$w_k(t)$	Time-dependent weight of trace k ; $= w_k(\tau)$, $\tau = t - t_k$ (see §4)

Identifiability. The trace decomposition is only meaningful if the ϕ_k are experimentally distinguishable; otherwise B collapses to one effective scalar per candidate.

4. Memory-Bias Decay

ANSATZ

For a trace k of age $\tau = t - t_k$, a minimal single-timescale exponential decay:

$$w_k(\tau) = s_k e^{-\tau / T_k}$$

Symbol	Meaning	Constraint
s_k	Salience amplitude at laydown	≥ 0
T_k	Decay timescale of trace k	> 0

One amplitude, one timescale — the most parsimonious falsifiable decay. Transient traces fade quickly (small T_k); strong anchors are long T_k . Re-access refreshes s_k and resets effective age. Limits: $\tau \rightarrow 0 \Rightarrow w_k \rightarrow s_k$; $\tau \rightarrow \infty \Rightarrow w_k \rightarrow 0$ (eventual forgetting preserved).

Deferred extension. A heavy-tailed (power-law) anchor component may be added *after* pilot data shows single-exponential decay underfits long-range anchor persistence. Held in a research note; excluded here to avoid over-parameterisation.

5. Recovering λ from Data — Falsifiability Spine

ANSATZ

Two-candidate log-odds **DERIVED** — linear in λ :

$$\ln \frac{P(y_i)}{P(y_j)} = \Delta U + \lambda \Delta B$$

$\Delta U = U(y_i) - U(y_j)$, $\Delta B = B(y_i; M_t) - B(y_j; M_t)$.

Falsifiability. A non-zero, stable slope across independent conditions is the positive result; a slope statistically indistinguishable from zero **falsifies the AIW claim in that regime**. $SE(\hat{\lambda}) \approx (1/|\Delta B|) \cdot \sqrt{(1 / [N \hat{p}_i(1-\hat{p}_i)])}$ — larger $|\Delta B|$ and N tighten the estimate.

Single-point estimate from $\hat{p}_i$:

$$\lambda^{\wedge} = \frac{\ln(\hat{p}_i / (1 - \hat{p}_i)) - \Delta U}{\Delta B}, \Delta B \neq 0$$

Least-squares fit (slope = $\hat{\lambda}$):

$$\lambda^{\wedge} = \frac{\sum_m \Delta B_m (\ell_m - \Delta U_m)}{\sum_m \Delta B_m^2}$$

$\ell_m = \ln(\hat{p}_{i,m} / (1 - \hat{p}_{i,m}))$

6. Consistency / Claim Boundaries

Constraint	Statement
No-signalling	For physical systems, B must be evaluated locally; bias cannot depend on spacelike-separated state.
Reduction	Recovers memoryless statistics at $\lambda = 0$ and where $\lambda \cdot \Delta B$ is small vs utility separations.
Prior art	Softmax, Bayesian updating, RL, Markov processes are not claimed as inventions. The claim is narrower: retained history acts as an active, measurable probabilistic weighting producing path-dependent divergence.
Status	λ , the decay law (§4), and WEL (§3) are testable hypotheses — they stand or fall on whether $\hat{\lambda}$ is reproducibly non-zero.

Legend (Symbols)

$P(y_i \mid S_t, O_t, M_t)$	Selection probability for candidate y_i
y_i	Candidate output / behavioural branch
S_t, O_t, M_t	State, observation/input, retained memory
$U(y_i; S_t, O_t)$	Present-state utility (memory-independent)
$B(y_i; M_t)$	Memory-derived bias term; encodes Active Information Weight (AIW)
λ	Memory-bias coupling coefficient
$\Delta B, \Delta U$	Bias / utility difference between candidates
ϕ_k, a_k, w_k	Trace alignment, salience, time-weight
$s_k, T_k, \tau, \hat{\lambda}$	Decay amplitude, timescale, trace age; empirical λ estimate